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Domain Specialization as the Key to Make Large Language Models Disruptive: A Comprehensive Survey

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Large language models (LLMs) have significantly advanced the field of natural language processing (NLP), providing a highly useful, task-agnostic foundation for a wide range of applications. However, directly applying LLMs to solve sophisticated problems in specific domains meets many hurdles, caused by the heterogeneity of domain data, the sophistication of domain knowledge, the uniqueness of domain objectives, and the diversity of the constraints (e.g., various social norms, cultural conformity, religious beliefs, and ethical standards in the domain applications). Domain specification techniques are key to make large language models disruptive in many

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applications. Specifically, to solve these hurdles, there has been a notable increase in research and practices conducted in recent years on the domain specialization of LLMs. This emerging field of study, with its substantial potential for impact, necessitates a comprehensive and systematic review to better summarize and guide ongoing work in this area. In this article, we present a comprehensive survey on domain specification techniques for large language models, an emerging direction critical for large language model applications. First, we propose a systematic taxonomy that categorizes the LLM domain-specialization techniques based on the accessibility to LLMs and summarizes the framework for all the subcategories as well as their relations and differences to each other. Second, we present an extensive taxonomy of critical application domains that can benefit dramatically from specialized LLMs, discussing their practical significance and open challenges. Last, we offer our insights into the current research status and future trends in this area.

Additional Key Words and Phrases: Large Language Models, Natural Language Processing, Domain Specialization

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1 INTRODUCTION

The evolution of natural language processing (NLP) and artificial intelligence (AI) models has witnessed a remarkable trajectory, beginning with the rule-based systems of the 1950s and 1960s, transitioning to statistical models in the 1990s, followed by the emergence of neural networks in the 2010s. Owing to the success of self-attention and Transformer-based neural network architecture [152], Pre-trained Language Models (PLMs) emerged and swiftly gained popularity in the late 2010s due to their ability to learn universal language representations from large-scale data in an unsupervised manner, which can be beneficial for many downstream NLP tasks such as commonsense reasoning [173], multiple-choice question answering [131], and story generation [13], while avoiding training new models from scratch. In the last few years, with the fast growth of large corpus and hardware capacities, researchers have found scaling up model and training data can continuously improve the model capacity, following the scaling law [65], eventually resulting in Large Language Models (LLMs) [166], such as GPT-3 [11] (175B parameters), PaLM [20] (540B parameters), and LLaMA [149] (65B parameters). LLMs, significantly outperforming smaller models in understanding and generating human-like text, have emerged as a promising AI research trend. Their potential to revolutionize natural and social sciences through efficient literature analysis, novel hypothesis generation, and complex data interpretation could accelerate research, enhance the discovery process, and facilitate interdisciplinary collaboration.

While LLMs hold great promise as general task solvers, effectively extending their functionality beyond mere "chatbot" roles poses significant challenges. This has led to the emergence of "domain specialization of LLMs". Specifically, **domain specialization** of Large Language Models (LLMs) is defined as *the process of customizing general-purpose LLMs according to specific domain contextual data, augmented by domain-specific knowledge, optimized by the domain's objective, and regulated by domain-specific constraints.* This shift towards domain specialization of LLMs is motivated by several compelling reasons. First, there are significant differences in conversation and language styles in different fields, roles, and tasks ranging from medical prescriptions to legal sentences, to online chatting, etc. The acquisition of such capabilities and experience even require human beings many years of training, a lot of which are hands-on and proprietary. Moreover, different fields, institutions, and teams have their own "business models" about which response will maximize their own utility function for their tasks, which is not directly replaceable by a single general-purpose LLMs solver with no customization. More importantly, the requirement of domain knowledge for professional-level

usage also need to be very in-depth, in-real-time, and accurate, none of which can be easily achieved by pre-trained LLMs. Many domain knowledge resources are proprietary assets and core competitiveness of the organizations that can never be leaked to general-purpose LLMs. Last but not the least, languages are constrained by social norms, cultural conformity, religious beliefs, legal requirements, and ethical practice, all of which are changing parameters in different locations, countries, populations, races, communities, etc., which make general-purpose LLMs impossible to be a one-fits-all solver without any customization.

Domain Specialization of LLMs is a critical yet challenging problem that requires inventing and integrating effective techniques to address the serious challenges. Particularly, there are three significant challenges.

Challenge 1: Difficulty keeping an LLM updated with the latest knowledge. The power of LLMs is attributed mainly to their massive training corpus. Yet, it also indicates LLMs tend to have a knowledge cut-off and lack sufficient access to the latest information, events, or discoveries. In many specialized domains, new discoveries, regulations, and best practices continuously emerge, making it difficult for LLMs to stay up-to-date. For instance, more than 30 thousand mainstream news articles are published every day [157]. For social media analysis and fact-checking, LLMs may not handle them since the knowledge extracted from the training corpus is offline. This indicates that regular re-training or continuous learning mechanisms are required to maintain LLMs' relevance and accuracy in these dynamic fields. However, ensuring the model freshness can be resource-intensive, as it necessitates continuous high-quality and up-to-date data collection, processing, and computationally intensive model re-training.

Challenge 2: Difficulty in learning all specialized knowledge of different domains in one LLM. LLMs, by default, possess general knowledge across a wide range of topics and may have seen and obtained specific knowledge for most domains. However, more popular or widely-discussed topics may be over-represented, while very domain-specific topics can usually be under-represented, which makes it difficult to be effectively learned for domain-specific tasks. In addition, domain-specific tasks often involve complex concepts, specialized terminology, and intricate relationships between entities. Without proper guidance, LLMs may generate plausible-sounding but inconsistent answers to similar queries (i.e., LLM's hallucination) or slightly rephrased questions [5]. This issue arises because LLMs are designed to predict the most likely word sequences based on the input rather than providing a definitive answer based on a structured knowledge base. Researchers have found users can guide the model to produce more relevant, accurate, and task-specific responses, enhancing the overall utility and effectiveness of AI systems across numerous domains by providing LLMs with a few task-specific demonstrations [166]. Nevertheless, providing LLMs with adequate demonstrations is not trivial since user instructions can often be vague, incomplete, or ambiguous, making it difficult to discern the intended meaning or desired outcome. Let alone LLMs tend to have a finite context window, typically determined by the maximum token length they can process (e.g., ChatGPT can only handle 4097 tokens).

Challenge 3: Intensive model and computational complexity required for downstream task learning. To better adapt to specific domain applications, downstream task learning is historically a commonly used practice to specialize language models. However, different from traditional language models, adapting an LLM to downstream tasks needs vast amounts of high-quality, task-specific data. Acquiring, cleaning, and pre-processing such data can be time-consuming and resource-intensive. Moreover, the sheer complexity of LLMs makes it challenging to identify the most appropriate down-stream task learning strategy, as the choice of hyperparameters, learning rate, and training duration can significantly impact the model's performance. Chen et al. [16] have also discussed down-stream task learning for LLMs may lead to severe *catastrophic forgetting* since the LLM with a complex architecture is more likely to forget previously learned knowledge and overfits to target domains. In addition to the data requirement and complex model architecture, LLMs typically consist of billions of parameters, e.g., both Generative Pre-trained Transformer 3 (GPT-3)